

TAMANNA

[Github](#) | [LinkedIn](#) | [Leetcode](#)

tamannas10110@gmail.com | +91-9877553894

AI Engineer with 5+ years designing and delivering production-grade Gen AI applications across cloud and enterprise environments — from architecture and prototyping through deployment and production operations. Specialized in building agentic AI systems and retrieval-augmented generation (RAG) workflows for enterprise document management, including document ingestion, processing, classification, and retrieval, while maintaining responsible AI practices and data security/compliance standards. Proficient in Python and React, with strong grounding in Gen AI frameworks (LangChain, LangGraph), prompt and context engineering, and software engineering best practices across the development lifecycle. Experienced integrating AI platforms with enterprise SaaS ecosystems via secure API-driven architectures, and working independently with general guidance to deliver scalable, reliable solutions adopted across large enterprise user bases.

WORK EXPERIENCE

Genpact (Senior Software Engineer – Agentic Systems)	Bangalore, Karnataka July 2024 - Present
• Owned complex engineering problems spanning multiple workflows and stakeholder teams — architected multi-agent orchestration for Genpact AP Trace using LangGraph, implementing supervisor/router patterns for autonomous reasoning and inter-agent coordination; flagged duplicates across 10,000+ invoices/month, reducing financial leakage risk by 15–20%. • Designed orchestration layers and backend services (Python, FastAPI) enabling multi-step agent execution and tool invocation for vendor onboarding and compliance validation workflows, applying prompt and context engineering — cutting manual processing effort by 40%. • Built RAG pipelines grounding LLM outputs in enterprise document data, applying evaluation practices to reduce hallucination and improve output reliability — treating agent outputs as production-critical, auditable behavior. • Implemented guardrails and monitoring for agent actions, supporting observability through logs and quality checks; led root-cause analysis and remediation for model/extraction quality regressions in production pipelines. • Owned a domain end-to-end — architecture, implementation, deployment, and operational health — for a Text-to-SQL AI backend service (Python, FastAPI) adopted by 200+ users across 10+ regions, reducing manual SQL dependency by 60%. • Applied secure coding and responsible AI practices — including output safety guardrails and compliance considerations — for GenAI document-processing pipelines (Azure OpenAI, Azure Document Intelligence) achieving 90%+ extraction accuracy across 500+ samples. • Established reusable design patterns and workflow artifacts for region-specific agent logic, evaluating fit-for-purpose patterns that accelerated onboarding by ~35% and cut rollout time from 6 weeks to under 4. • Integrated enterprise platforms (ServiceNow, Salesforce) via secure, authenticated REST APIs with source-controlled development practices; led through influence across global, cross-functional teams — aligning stakeholders and translating complex Gen AI/agentic concepts into business value.	
CGI (Software Engineer)	Bangalore, Karnataka June 2021 – June 2024
• Owned backend services end-to-end — design, implementation, deployment, and operational health — for cloud-native enterprise applications (React.js, Python — Flask/FastAPI), supporting 200+ enterprise users. • Designed secure, scalable REST API architectures (FastAPI, PostgreSQL, Redis) with a focus on reliability and performance, reducing average API response time by 35%. • Built Python-based automations and integrations, converting recurring operational issues into engineered, maintainable workflow solutions that replaced manual, legacy enterprise processes — reducing processing effort by 40–50% and eliminating 15–20 hrs/week of manual work. • Developed reusable React component libraries and coding standards, raising code quality and consistency across the team's solutions — cutting feature delivery time by 30%. • Built Selenium-based automation scripts for critical enterprise workflows, reducing failures by 30%; maintained technical documentation and used source control for collaborative development, code review, and debugging. • Operated as a fully independent practitioner across the full application lifecycle with minimal supervision — requirements, architecture, build, and deployment — applying strong analytical and problem-solving skills across legacy systems and ambiguous environments. • Drove alignment across dependencies and stakeholders — acted as a technical interface to business teams, leading requirements gathering and demos, and clearly communicating technical concepts to non-technical audiences across cross-functional teams. • Worked across Linux and Windows environments using Python and command-line scripting for builds, deployments, and debugging, contributing to engineering best practices and operational reliability.	

EDUCATION

Qualification	Institute	CGPA / Percentage	Time of Completion
Bachelor of Technology	NIT Jalandhar	CGPA – 7.92/10	August 2017 – May 2021
Class XII	GGN Public School, Ludhiana	91.8 %	April 2016 – March 2017
Class X	GGN Public School, Ludhiana	CGPA – 10/10	April 2014 – March 2015

TECHNICAL SKILLS

- **AI / GenAI / Agentic AI:** LLMs, Generative AI, Agentic AI, Multi-Agent Orchestration, LangChain, LangGraph, RAG, Prompt Engineering, Azure OpenAI, Azure Document Intelligence, Vector Databases (FAISS/Pinecone/ChromaDB), Function/Tool Calling, Scikit-Learn, TensorFlow
- **Languages:** Python, JavaScript, SQL, C++
- **Frontend:** React.js, Redux Toolkit, MUI (Material-UI), Tailwind CSS, HTML5/CSS3
- **Backend & APIs:** FastAPI, Flask, REST APIs, WebSockets, Microservices Architecture
- **Databases & Caching:** PostgreSQL, Redis, MongoDB
- **Cloud & DevOps:** Azure (App Service, Functions, OpenAI Service), Docker, Git, CI/CD
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Automation & Testing:** Selenium, PyTest
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, System Design

PERSONAL PROJECT

Agentic Medical Appointments Processing – [DEMO](#)

The system simulates how medical appointments move through an AI-orchestrated pipeline:

- **Orchestrator** — A master agent reads appointments from a database and ranks them by dynamic priority (client, specialty, notes).
- **Stage agents** — Each appointment passes through 6 configurable stages (e.g. Intake, Insurance, Clinical Review). Each stage is an LLM-backed agent that returns a standardized outcome.
- **Exception queue** — If a stage returns Escalate, the case goes to a human concierge.
- **Human resolution** — A human resolves the issue; the appointment becomes Cleared and the workflow resumes automatically.

Tech Stack Used – React, Vite, FastAPI, Python, SQLite, LangChain, LangGraph, Hugging Face LLM

CERTIFICATIONS

- [Document AI: From OCR to Agentic Doc Extraction](#) – May 2026
- [Databricks Certified Generative AI Engineer Associate](#)– July 2025